

## FITCHBURG, MA, USA

WMO: 725107

Lat: 42.552N

Lon: 71.756W

Elev: 106

StdP: 100.06

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04780

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.5      | -13.6 | -24.4                         | 0.4 | -14.2 | -22.0 | 0.5 | -11.7 | 11.3                  | -4.0 | 10.2 | -4.7 | 3.1                   | 310  | 0.498 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.9                   | 32.7            | 22.6 | 31.2 | 21.8 | 29.7 | 20.9 | 24.2                | 29.6 | 23.4 | 28.4 | 22.6 | 27.3 | 4.2                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.6                            | 17.6 | 26.5 | 22.0 | 16.9 | 26.0 | 21.1 | 15.9 | 25.4 | 73.2          | 29.7 | 69.9 | 28.2 | 66.9 | 27.4 | 27.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  |    |    |
| 9.3               | 8.3  | 7.3 | -20.1                      | 35.5 | 2.6                | 1.7 | -22.0                                              | 36.8 | -23.5      | 37.8 | -25.0      | 38.8 | -26.8      | 40.0 |    |    |
|                   |      |     | -20.5                      | 25.6 | 2.6                | 0.8 | -22.4                                              | 26.2 | -23.9      | 26.7 | -25.3      | 27.2 | -27.2      | 27.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.0   | -3.4  | -2.2  | 2.3   | 9.1   | 14.9  | 19.8  | 22.9  | 22.0  | 18.0  | 11.0  | 5.3   | -0.3 |
|                                                                                  | DBStd       | 10.25  | 5.85  | 5.30  | 5.20  | 4.82  | 4.48  | 3.97  | 3.17  | 3.14  | 3.97  | 4.52  | 4.71  | 5.02 |
|                                                                                  | HDD10.0     | 1600   | 417   | 342   | 244   | 74    | 7     | 0     | 0     | 0     | 0     | 42    | 154   | 319  |
|                                                                                  | HDD18.3     | 3447   | 675   | 574   | 497   | 282   | 129   | 29    | 3     | 6     | 54    | 229   | 392   | 576  |
|                                                                                  | CDD10.0     | 1603   | 1     | 1     | 7     | 46    | 157   | 293   | 401   | 371   | 239   | 74    | 13    | 2    |
|                                                                                  | CDD18.3     | 408    | 0     | 0     | 1     | 3     | 21    | 72    | 146   | 118   | 44    | 3     | 0     | 0    |
|                                                                                  | CDH23.3     | 3843   | 0     | 1     | 4     | 61    | 267   | 695   | 1375  | 1029  | 379   | 33    | 0     | 0    |
|                                                                                  | CDH26.7     | 1285   | 0     | 0     | 1     | 19    | 88    | 231   | 509   | 324   | 109   | 3     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.8    | 3.1   | 3.2   | 3.4   | 3.3   | 2.8   | 2.6   | 2.5   | 2.3   | 2.2   | 2.5   | 2.7   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 1148   | 87    | 83    | 100   | 88    | 89    | 97    | 97    | 96    | 97    | 102   | 95    | 105  |
|                                                                                  | PrecMax     | 1520   | 168   | 271   | 263   | 204   | 178   | 224   | 221   | 225   | 219   | 340   | 173   | 191  |
|                                                                                  | PrecMin     | 832    | 30    | 36    | 15    | 15    | 21    | 16    | 38    | 13    | 20    | 3     | 27    | 34   |
|                                                                                  | PrecStd     | 173    | 33    | 42    | 49    | 52    | 39    | 54    | 46    | 51    | 57    | 75    | 33    | 33   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.0  | 16.7  | 22.3  | 29.4  | 32.4  | 33.7  | 34.8  | 33.8  | 32.5  | 26.5  | 20.9  | 17.4 |
|                                                                                  |             | MCWB   | 12.9  | 11.6  | 13.5  | 16.8  | 19.5  | 22.9  | 23.8  | 23.8  | 22.7  | 18.2  | 15.3  | 13.8 |
|                                                                                  | 2%          | DB     | 10.9  | 11.9  | 17.1  | 24.2  | 29.1  | 31.4  | 32.8  | 31.9  | 29.7  | 23.8  | 18.1  | 12.6 |
|                                                                                  |             | MCWB   | 7.7   | 7.9   | 10.9  | 13.7  | 18.4  | 21.7  | 22.9  | 22.6  | 21.2  | 17.9  | 13.7  | 9.8  |
|                                                                                  | 5%          | DB     | 7.2   | 8.5   | 13.3  | 21.1  | 26.3  | 29.2  | 31.2  | 30.0  | 27.4  | 21.5  | 16.0  | 9.8  |
|                                                                                  |             | MCWB   | 4.5   | 4.6   | 7.7   | 12.6  | 16.8  | 20.2  | 22.1  | 21.4  | 19.8  | 16.6  | 11.5  | 6.8  |
|                                                                                  | 10%         | DB     | 4.2   | 5.5   | 10.3  | 18.0  | 23.6  | 27.4  | 29.6  | 28.3  | 25.2  | 18.9  | 13.4  | 7.2  |
|                                                                                  |             | MCWB   | 1.6   | 2.2   | 5.9   | 11.0  | 15.7  | 19.3  | 21.2  | 20.7  | 19.0  | 14.3  | 9.8   | 4.5  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 13.3  | 13.5  | 14.9  | 18.5  | 21.5  | 24.3  | 25.7  | 25.1  | 24.0  | 21.1  | 17.4  | 14.3 |
|                                                                                  |             | MCDB   | 14.6  | 16.0  | 20.1  | 25.0  | 29.4  | 30.9  | 32.2  | 31.1  | 29.8  | 23.7  | 19.6  | 16.2 |
|                                                                                  | 2%          | WB     | 8.1   | 8.4   | 11.9  | 16.1  | 20.1  | 23.1  | 24.4  | 24.0  | 22.5  | 19.2  | 14.7  | 10.4 |
|                                                                                  |             | MCDB   | 9.9   | 11.3  | 16.4  | 22.7  | 26.4  | 28.7  | 30.0  | 28.9  | 27.3  | 22.3  | 17.1  | 12.3 |
|                                                                                  | 5%          | WB     | 4.6   | 4.7   | 8.7   | 14.1  | 18.6  | 22.1  | 23.4  | 23.2  | 21.6  | 17.0  | 12.4  | 7.4  |
|                                                                                  |             | MCDB   | 7.3   | 7.9   | 12.5  | 19.4  | 24.0  | 27.1  | 28.7  | 27.5  | 25.5  | 20.8  | 14.8  | 9.1  |
|                                                                                  | 10%         | WB     | 2.0   | 2.5   | 6.2   | 11.7  | 17.0  | 20.8  | 22.6  | 22.3  | 20.5  | 15.2  | 10.2  | 4.7  |
|                                                                                  |             | MCDB   | 3.8   | 5.1   | 9.8   | 16.8  | 21.8  | 25.4  | 27.5  | 26.5  | 24.0  | 18.3  | 13.3  | 6.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 9.1   | 10.0  | 10.1  | 11.6  | 11.6  | 10.9  | 10.9  | 11.0  | 11.3  | 10.9  | 10.0  | 8.7  |
|                                                                                  |             | MCDBR  | 11.6  | 13.8  | 14.9  | 16.8  | 15.9  | 13.1  | 12.3  | 12.2  | 13.1  | 13.9  | 13.4  | 12.0 |
|                                                                                  |             | MCWBR  | 9.7   | 10.2  | 9.4   | 8.6   | 7.4   | 5.7   | 4.8   | 4.9   | 6.2   | 8.0   | 9.3   | 10.0 |
|                                                                                  | 5% WB       | MCDBR  | 11.3  | 12.7  | 13.4  | 14.6  | 13.4  | 11.6  | 10.9  | 10.2  | 10.8  | 11.7  | 11.2  | 11.4 |
|                                                                                  |             | MCWBR  | 9.8   | 10.1  | 9.6   | 8.8   | 7.2   | 5.8   | 5.0   | 5.0   | 6.1   | 8.1   | 9.3   | 10.4 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.284  | 0.295 | 0.324 | 0.366 | 0.402 | 0.441 | 0.461 | 0.428 | 0.372 | 0.331 | 0.305 | 0.286 |      |
|                                                                                  | taud        | 2.456  | 2.436 | 2.448 | 2.379 | 2.315 | 2.233 | 2.185 | 2.262 | 2.397 | 2.530 | 2.556 | 2.545 |      |
|                                                                                  | Ebn at Noon | 866    | 913   | 924   | 902   | 874   | 837   | 814   | 831   | 858   | 862   | 838   | 838   |      |
|                                                                                  | Edh at Noon | 74     | 89    | 100   | 115   | 126   | 138   | 143   | 128   | 104   | 81    | 68    | 63    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.79   | 2.65  | 3.65  | 4.63  | 5.10  | 5.52  | 5.84  | 5.25  | 4.25  | 2.76  | 1.92  | 1.46  |      |
|                                                                                  | RadStd      | 0.11   | 0.21  | 0.25  | 0.50  | 0.61  | 0.54  | 0.36  | 0.26  | 0.30  | 0.26  | 0.16  | 0.15  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +135    | +64     |
| (47) | Regional (77 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +90     | +39     |

Nomenclature: See separate page